

Probability

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A probability question

Problem

A bag contains three red balls and two blue balls. One ball is selected, with each ball equally likely to be chosen.

What is the probability that the selected ball is red?

Outcomes and events

A **sample space** Ω is the set of possible outcomes. An **outcome** is an element $\omega \in \Omega$. An **event** is a subset of Ω . Label the red balls r_1, r_2, r_3 and the blue balls b_1, b_2 .

$$\Omega = \{r_1, r_2, r_3, b_1, b_2\}.$$

The event “the selected ball is red” is

$$A = \{r_1, r_2, r_3\}.$$

The event A occurs exactly when the observed outcome belongs to A .

Probability

For a finite or countably infinite sample space Ω , a **probability measure** assigns a number $P(A)$ to every event $A \subseteq \Omega$, satisfying:

1. $P(A) \geq 0$ for every event A .
2. $P(\Omega) = 1$.
3. For pairwise disjoint events $A_1, A_2, \dots$,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

Pairwise disjoint means $A_i \cap A_j = \emptyset$ whenever $i \neq j$. These are the **probability axioms**. Other rules, such as the addition rule, can be proved from them.

Proving the addition rule

For events A, B , we will prove

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Let $C = B \setminus A$: the outcomes in B but not in A .

$A \cup B$ is the union of the disjoint events A and C . By additivity,

$$P(A \cup B) = P(A) + P(C).$$

B is the union of the disjoint events $A \cap B$ and C . Thus

$$P(B) = P(A \cap B) + P(C).$$

Substitute $P(C) = P(B) - P(A \cap B)$ into the first equality to obtain the addition rule.

Specifying a discrete model

We can specify the model by giving each outcome a probability $p(\omega) = P(\{\omega\})$, with

$$p(\omega) \geq 0, \quad \sum_{\omega \in \Omega} p(\omega) = 1.$$

Because distinct outcomes are disjoint events,

$$P(A) = \sum_{\omega \in A} p(\omega).$$

The axioms constrain a model; they do not tell us which model fits a particular experiment.

Selecting a red ball: solution

Sample space. Label the individual balls:

$$\Omega = \{r_1, r_2, r_3, b_1, b_2\}.$$

Outcome probabilities. Each ball is equally likely:

$$p(\omega) = \frac{1}{5} \quad (\omega \in \Omega).$$

Target event. “The selected ball is red” means $A = \{r_1, r_2, r_3\}$.

Calculation.

$$P(A) = \sum_{\omega \in A} p(\omega) = \frac{1}{5} + \frac{1}{5} + \frac{1}{5} = \frac{3}{5}.$$

Equally likely outcomes

Suppose Ω is finite and all its outcomes are equally likely. Each outcome must then have probability $1/|\Omega|$, where $|\Omega|$ is the number of outcomes. For any event A ,

$$P(A) = \frac{|A|}{|\Omega|}.$$

This is where counting becomes useful: count the favorable outcomes and all possible outcomes.

The formula requires equally likely outcomes. It is not the definition of probability in every model.

Two dice

Problem

A fair red die and a fair blue die are rolled independently.

What is the probability that their sum is eight?

Two dice: solution

Sample space. Record the red and blue results as (i, j) :

$$\Omega = \{(i, j) : i, j \in \{1, 2, \dots, 6\}\}, \quad |\Omega| = 36.$$

Outcome probabilities. Fair, independent rolls give $p((i, j)) = 1/36$ for every $(i, j) \in \Omega$.

Target event. The sum is eight:

$$\begin{aligned} A &= \{(i, j) \in \Omega : i + j = 8\} \\ &= \{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\}. \end{aligned}$$

Calculation. $P(A) = \sum_{\omega \in A} p(\omega) = 5/36$.

Event notation

For events $A, B \subseteq \Omega$:

Event	Meaning
$A \cap B$	Both A and B occur.
$A \cup B$	At least one of A, B occurs.
$A^c = \Omega \setminus A$	A does not occur.
$\emptyset$	No outcome: the impossible event.
Ω	Every outcome: the certain event.

The union uses inclusive “or”: both events may occur.

The complement rule

An event A and its complement A^c are disjoint, and $A \cup A^c = \Omega$. By additivity,

$$1 = P(\Omega) = P(A) + P(A^c).$$

Therefore

$$\boxed{P(A^c) = 1 - P(A).}$$

In particular, $P(\emptyset) = 0$ and $0 \leq P(A) \leq 1$. Sometimes it is easier to count the outcomes where an event *does not* occur.

At least one head

Problem

A fair coin is flipped four times independently.

What is the probability of getting at least one head?

At least one head: solution

Sample space and probabilities. Write H for heads, T for tails, and x_i for flip i :

$$\Omega = \{(x_1, x_2, x_3, x_4) : x_i \in \{H, T\}\}, \quad p(\omega) = \frac{1}{16} \quad (\omega \in \Omega).$$

Target event. At least one flip is heads:

$$A = \{(x_1, x_2, x_3, x_4) \in \Omega : \text{some } x_i = H\}.$$

Calculation. The complement contains just one outcome:

$$A^c = \{(T, T, T, T)\}.$$

Hence

$$P(A) = 1 - P(A^c) = 1 - \frac{1}{16} = \frac{15}{16}.$$

“At least one” often becomes simpler after taking a complement.

The addition rule

Problem

A fair die is rolled once.

What is the probability that the result is even or at least four? Here “or” includes the possibility that both conditions hold.

The addition rule: solution

Sample space and probabilities.

$$\Omega = \{1, 2, 3, 4, 5, 6\}, \quad p(\omega) = 1/6 \quad (\omega \in \Omega).$$

Target event. Let A mean even and B mean at least four:

$$A = \{2, 4, 6\}, \quad B = \{4, 5, 6\}, \quad A \cup B = \{2, 4, 5, 6\}.$$

Calculation. We seek $P(A \cup B)$:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Since $A \cap B = \{4, 6\}$,

$$P(A \cup B) = \frac{3}{6} + \frac{3}{6} - \frac{2}{6} = \frac{4}{6}.$$

If A and B are disjoint, the overlap term is zero.

Sampling without replacement

Problem

A bag contains three red balls and two blue balls. Two balls are selected without replacement, with each pair of distinct balls equally likely.

What is the probability that the two balls have the same color?

Sampling without replacement: solution

Sample space and probabilities. Let $R = \{r_1, r_2, r_3\}$, $B = \{b_1, b_2\}$, and $L = R \cup B$. Choose two distinct balls uniformly, ignoring their order.

$$\Omega = \{U \subseteq L : |U| = 2\}, \quad p(U) = \frac{1}{\binom{5}{2}} = \frac{1}{10} \quad (U \in \Omega).$$

Target event. The colors match:

$$\begin{aligned} M &= \{U \in \Omega : U \subseteq R \text{ or } U \subseteq B\} \\ &= \{\{r_1, r_2\}, \{r_1, r_3\}, \{r_2, r_3\}, \{b_1, b_2\}\}. \end{aligned}$$

Calculation.

$$P(M) = \frac{\binom{3}{2} + \binom{2}{2}}{\binom{5}{2}} = \frac{4}{10} = \frac{2}{5}.$$

Conditional probability

For events A, B with $P(A) > 0$, define

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}.$$

We keep only outcomes in A , then divide by $P(A)$ so that the probabilities within A sum to 1.

$$P(A \mid A) = 1.$$

In a finite uniform model this becomes

$$P(B \mid A) = \frac{|A \cap B|}{|A|}.$$

The denominator is the event we are *given*.

The order matters

Problem

A fair coin is flipped twice independently. You are told that both flips are heads.

What is the probability that the first flip is heads?

The order matters: solution

Sample space and probabilities.

$$\Omega = \{HH, HT, TH, TT\}, \quad p(\omega) = \frac{1}{4} \quad (\omega \in \Omega).$$

Events. The given event is $B = \{HH\}$ (both heads). The target event is $A = \{HH, HT\}$ (first heads).

Calculation. Since $A \cap B = B$,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{1/4}{1/4} = 1.$$

Reversing the given and target events gives a different probability:

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{1/4}{1/2} = \frac{1}{2} \neq 1.$$

The conditioning event and the target event have exchanged roles.

The multiplication rule

Problem

A bag contains three red balls and two blue balls. Draw one ball uniformly, keep it out of the bag, and draw a second ball uniformly from those remaining.

What is the probability that both balls are red?

The multiplication rule: solution

Sample space and probabilities. Let $R = \{r_1, r_2, r_3\}$ and $L = \{r_1, r_2, r_3, b_1, b_2\}$. Draw two balls uniformly without replacement, recording their order:

$$\Omega = \{(u, v) \in L^2 : u \neq v\}, \quad p((u, v)) = \frac{1}{20}.$$

Here $L^2 = L \times L$, and u, v are the first and second balls.

$$R_1 = \{(u, v) \in \Omega : u \in R\}, \quad R_2 = \{(u, v) \in \Omega : v \in R\}.$$

Target event. Two red balls means $R_1 \cap R_2$.

Calculation. Use $P(A \cap B) = P(A)P(B \mid A)$:

$$P(R_1 \cap R_2) = \frac{3}{5} \cdot \frac{2}{4} = \frac{3}{10}.$$

After a red ball is removed, two of the four remaining balls are red.

Independence

Events A and B are **independent** if

$$P(A \cap B) = P(A)P(B).$$

If $P(A) > 0$, this is equivalent to

$$P(B \mid A) = P(B).$$

Knowing that A occurred does not change the probability of B .
Independence is a property of events in a probability model.

Replacement changes the model

Problem

Draw twice from a bag of three red balls and two blue balls, choosing uniformly at each draw. The first ball is red.

What is the probability that the second ball is red

- ▶ if the first ball is replaced and the bag is mixed?
- ▶ if the first ball is not replaced?

Replacement: solution

Let $L = \{r_1, r_2, r_3, b_1, b_2\}$ and $R = \{r_1, r_2, r_3\}$. **With replacement:**

$$\Omega_{\text{with}} = L^2, \quad p_{\text{with}}((u, v)) = \frac{1}{25}.$$

Without replacement:

$$\Omega_{\text{without}} = \{(u, v) \in L^2 : u \neq v\}, \quad p_{\text{without}}((u, v)) = \frac{1}{20}.$$

Events. In each model, the given and target events are

$$R_1 = \{(u, v) \in \Omega : u \in R\}, \quad R_2 = \{(u, v) \in \Omega : v \in R\}.$$

Calculation. Using the probability measure of each model,

$$P(R_2 \mid R_1) = \begin{cases} 3/5 = P(R_2) & \text{with replacement,} \\ 2/4 \neq P(R_2) = 3/5 & \text{without replacement.} \end{cases}$$

Disjoint does not mean independent

Problem

A fair die is rolled once.

Are the events “the result is one” and “the result is two” independent?
They cannot both occur. Does that imply independence?

Disjoint events: solution

Sample space and probabilities. $\Omega = \{1, 2, 3, 4, 5, 6\}$ and $p(\omega) = 1/6$ for every $\omega \in \Omega$.

Events. Let $A = \{1\}$ and $B = \{2\}$. We compare $P(A \cap B)$ with $P(A)P(B)$, and $P(B | A)$ with $P(B)$. These events are disjoint, but

$$P(A \cap B) = 0 \neq \frac{1}{36} = P(A)P(B).$$

If A occurs, B becomes impossible:

$$P(B | A) = 0 \neq \frac{1}{6} = P(B).$$

These events are dependent. Knowing that the result is one changes the probability of getting two from $1/6$ to 0.

A partition of the sample space

Events $A_1, \dots, A_m$ form a **partition** of Ω if they are pairwise disjoint and their union is Ω . Every outcome belongs to exactly one of these events. For any event B , its pieces

$$B \cap A_1, B \cap A_2, \dots, B \cap A_m$$

are disjoint and together make up B .

$$P(B) = \sum_{i=1}^m P(B \cap A_i).$$

The law of total probability

Let $A_1, \dots, A_m$ partition Ω , with $P(A_i) > 0$. Apply the multiplication rule to each piece:

$$\begin{aligned} P(B) &= \sum_{i=1}^m P(B \cap A_i) \\ &= \boxed{\sum_{i=1}^m P(B \mid A_i)P(A_i)}. \end{aligned}$$

In particular, if $0 < P(A) < 1$,

$$P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c).$$

We average the conditional probabilities, weighted by how likely each case is.

A spam detector

Problem

In a hypothetical email population, one percent of emails are spam. A detector flags ninety percent of spam emails, but it also flags five percent of non-spam emails.

An email is selected at random and is found to be flagged. What is the probability that it is spam?

A spam detector: solution

Sample space. Record an email as (s, f) : $s = 1$ means actually spam, and $f = 1$ means flagged by the detector. In each coordinate, 0 means no.

$$\Omega = \{(0, 0), (0, 1), (1, 0), (1, 1)\}.$$

Events. Actual spam and being flagged are different events:

$$S = \{(1, 0), (1, 1)\}, \quad F = \{(0, 1), (1, 1)\}.$$

S includes spam that the detector misses, namely $(1, 0)$.

Given information. The problem states

$$P(S) = 0.01, \quad P(F \mid S) = 0.90, \quad P(F \mid S^c) = 0.05.$$

We are told that F occurred. We want $P(S \mid F)$.

A spam detector: outcome probabilities

The outcome $(1, 1)$ means *both* actual spam and flagged:

$$\{(1, 1)\} = S \cap F.$$

By the multiplication rule,

$$\begin{aligned} p((1, 1)) &= P(S \cap F) = P(S)P(F \mid S) \\ &= 0.01 \times 0.90 = 0.009. \end{aligned}$$

Similarly, the remaining outcome probabilities are

ω	Event	$p(\omega)$
$(1, 0)$	$S \cap F^c$	$0.01(1 - 0.90) = 0.001$
$(0, 1)$	$S^c \cap F$	$0.99(0.05) = 0.0495$
$(0, 0)$	$S^c \cap F^c$	$0.99(1 - 0.05) = 0.9405$

These probabilities follow from the stated rates. We have not assumed that S and F are independent.

A spam detector: conditional probability

Given event. The email is flagged:

$$F = \{(0, 1), (1, 1)\}.$$

Only these two outcomes remain possible, but they are not equally likely.

Target within the given event. Actual spam is $S = \{(1, 0), (1, 1)\}$, so

$$S \cap F = \{(1, 1)\}.$$

Calculation. Add the probabilities of all flagged outcomes in the denominator:

$$\begin{aligned} P(S \mid F) &= \frac{P(S \cap F)}{P(F)} = \frac{p((1, 1))}{p((0, 1)) + p((1, 1))} \\ &= \frac{0.009}{0.0495 + 0.009} \approx 0.154 = 15.4\%. \end{aligned}$$

Bayes' theorem

Let A, B be events with $P(A) > 0$ and $P(B) > 0$. The same intersection can be written in two ways:

$$P(A \cap B) = P(B | A)P(A) = P(A | B)P(B).$$

Divide by $P(B)$:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}.$$

This is **Bayes' theorem**. It relates the two directions of conditional probability.

$P(A)$ is the **prior** probability; $P(A | B)$ is the **posterior** probability after learning B .

Applying Bayes' theorem

In the four-outcome model, $\Omega = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$, with probabilities 0.9405, 0.0495, 0.001, 0.009, respectively.

Given $F = \{(0, 1), (1, 1)\}$, we seek $P(S \mid F)$ for $S = \{(1, 0), (1, 1)\}$. Total probability gives

$$\begin{aligned}P(F) &= P(F \mid S)P(S) + P(F \mid S^c)P(S^c) \\&= 0.90(0.01) + 0.05(0.99) = 0.0585.\end{aligned}$$

Bayes' theorem now gives

$$P(S \mid F) = \frac{0.90(0.01)}{0.0585} = \frac{0.009}{0.0585} \approx 15.4\%.$$

Most emails are not spam. Even a small false-alarm rate can therefore produce many flagged non-spam emails.

The prior matters

Problem

Suppose half the emails are now spam. The detector still flags ninety percent of spam and five percent of non-spam emails.

For a randomly selected email that is flagged, what is the probability that it is spam now?

The prior matters: solution

Sample space. Use $\Omega = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$, with coordinates (spam, flagged), where 1 means yes and 0 means no.

Outcome probabilities. The new rates give

ω	(0, 0)	(0, 1)	(1, 0)	(1, 1)
$p(\omega)$	0.475	0.025	0.05	0.45

Events. Given $F = \{(0, 1), (1, 1)\}$, the target is $S = \{(1, 0), (1, 1)\}$.

Thus

$$P(S \mid F) = \frac{0.90(0.5)}{0.90(0.5) + 0.05(0.5)} = \frac{18}{19} \approx 94.7\%.$$

The same conditional detector rates gave 15.4% when spam was rare, and 94.7% when spam was common.

A detector's output cannot be interpreted from its detection rate alone.

Probability rules

Complement $P(A^c) = 1 - P(A)$

Union $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Conditional $P(B | A) = P(A \cap B)/P(A)$

Independent $P(A \cap B) = P(A)P(B)$

Bayes $P(A | B) = P(B | A)P(A)/P(B)$

Conditional probabilities here require a positive-probability conditioning event. Start by specifying the outcomes, their probabilities, and the event being asked about. Then choose the appropriate rule.